

Jimmy Yao

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EXPERIENCE

Epic Systems

Software Developer (AI/ML)

Verona, WI

Aug 2023 – Present

- Led development of referral-focused LLM workflow for clinical chart review, translating clinician interviews into chart-context design, prompts, orchestration logic, generated outputs, and review flows
- Built core pieces of production LLM summary workflows used across 100+ hospital systems, transforming long patient chart histories into problem-oriented outputs with traceable chart evidence
- Built agent-assisted prompt evaluation workflow for model upgrades, running clinical examples and edge cases through candidate prompts/models to measure F1, inspect failure modes, and test output stability against clinician-oriented inclusion criteria
- Assembled AI-ready chart context from fragmented patient data across clinical workflow, external-record exchange, orders, labs, and imaging systems
- Helped develop synthetic patient generation systems for safer testing of clinical AI workflows
- Partnered with clinicians to refine generated summary structure, chart-context selection, and review flows based on how providers prepare for patient visits
- Built scheduling and prior-authorization workflow tools across 10+ hospital systems, including rule-based warnings, authorization timing estimates, and pattern-mining workflows for recurring unnecessary approval requirements

M&T Bank

Software Engineer Intern

Buffalo, NY

Jun 2022 – Dec 2022

- Developed UX interview analysis pipeline using AWS Transcribe, BERT/T5 summarization, sentiment analysis, and topic clustering to turn raw interviews into structured research insights
- Designed React dashboard for product managers and UX researchers to review sentiment trends, BERTopic clusters, and generated summaries across interview sessions
- Deployed prototype through GitLab CI/CD on AWS infrastructure, including CDK, S3, Lambda, and DynamoDB

AIH, LLC

Software Engineer

Rochester, NY

Mar 2021 – Jun 2022

- Prototyped multimodal stroke-screening application using Flutter and TensorFlow Lite, combining speech-audio and facial-image models to detect stroke indicators with 76% accuracy on evaluation data
- Owned backend infrastructure for health-data processing, building Python Flask APIs with Docker, InfluxDB, MongoDB, and AWS EC2; introduced Docker Compose for local development
- Maintained Kotlin Android applications for Bluetooth wearable data, using threshold detection and moving averages to surface posture-monitoring insights
- Optimized InfluxDB queries and indexing, improving API response times by 3× for time-series health data

AIH, LLC

Android Developer

Rochester, NY

Jan 2020 – Mar 2021

- Developed aiSpine Android application for collecting and displaying physiological parameters from Bluetooth smart wearable devices
- Integrated Auth0 authentication for secure user access in a consumer health-monitoring application

SELECTED PROJECTS

Meridian CLI | *Python, Rust, MCP, FastMCP*

Feb 2026 – Present

- Built agent orchestration CLI for decomposing complex workflows across specialized agents, models, and tool harnesses, with isolated context, resumable execution, and checkpointed reports
- Implemented delegated-agent lifecycle across foreground/background spawns, continuation/forking, parent-child tracking, cancellation, changed-file inspection, reports, and spawn statistics
- Designed agent-profile system for repeatable multi-model workflows, including model policies, harness fallback chains, skill injection, tool permissions, sandbox/approval settings, and primary/subagent modes
- Built file-backed memory layer separating durable knowledge-base content, active work-item memory, archived work history, and local runtime state, with Git autosync for shared project context
- Built [jupyter-workbench](#), a CLI/MCP package that gives LLM agents persistent Jupyter sessions, notebook lineage, and 3D visualization workflows through PyVista and trame
- Created reusable agent workflows for [software engineering](#), [micro-CT image analysis](#), and [creative writing](#), with the creative-writing workflow earning 150+ GitHub stars

Mars Agents | *Rust, Python, npm, Git*

Mar 2026 – Present

- Created package manager that installs reusable agent profiles, skills, MCP servers, hooks, and model aliases from Git/local sources into Claude Code, Codex, Cursor, and OpenCode

- Engineered deterministic sync pipeline that resolves dependency graphs before file mutation, applies filters and renames, detects collisions, writes atomically, syncs linked targets, and persists lock-file state
- Added package workflows for install, upgrade, adoption, target linking, health checks, and repair, with managed-file tracking to avoid overwriting user-owned files

MorphoLens | *React, Python, Gemini 3 Pro*

Dec 2025

- Developed research-analysis tool that converts domain-expert queries into executable Python analysis with browser-based NumPy, OpenCV, and scikit-image workflows
- Composed shareable role packages that bundle prompts, analysis code, and execution patterns for reproducible scientific workflows
- Rendered masks, plots, and measurement overlays on source images through an interactive canvas while generated Python code executed

Meridian Flow | *Go, React/TypeScript, PostgreSQL*

Oct 2025 – Present

- Created collaborative AI writing environment with rich document editing, local-first IndexedDB sync, and persistent server-side streaming generation
- Tested RAG and agentic file-exploration approaches for long-context writing, using real writing projects to evaluate retrieval, context assembly, and AI-assisted revision patterns

OPEN SOURCE AND SMALL PROJECTS

- **LlamaIndex EPUB parser**: Added .epub parser to the LlamaIndex/LlamaHub ingestion ecosystem using EbookLib and html2text
- **Kaggle CLI auth fix**: Fixed access-token authentication failure when KAGGLE_API_TOKEN is unset
- **EpubSplit performance fix**: Optimized EPUB splitting path, reducing large split runtime from an incomplete 22+ minute run to roughly 1.5 minutes in benchmarked CLI tests
- **maxent-ai/converse NaN fix**: Patched semantic text segmentation failure when utterances containing only stop words produced NaN embeddings
- **repo-viewer**: Created mobile-friendly read-only repository browser with Markdown/Mermaid preview, image galleries, clipboard uploads, Tailscale Serve integration, and optional Basic Auth
- **ebook-tree-parser**: Published Python package for parsing EPUB table-of-contents trees with token-count estimates for long-document workflows
- **SAMGIT**: Developed reverse diffusion pipeline combining Segment Anything, image-to-text transformers, Stable Diffusion inversion, and CLIP-based reconstruction evaluation
- **Flow-Style-VTON**: Extended virtual try-on pipeline with neural style transfer variants before warp, on warped garments, and after generated try-on; evaluated outputs with FID and SSIM
- **gdocwebnovelsync**: Automated Google Docs-to-webnovel publishing across multiple platforms through Google Docs API integration and browser automation
- **RoyalRoad GPT-2 experiment**: Scraped 5,000 fiction synopses and fine-tuned GPT-2 355M in Colab for early generative-writing experiments
- **Frontiers in Immunology co-author**: Supported breast-cancer immunotherapy study through wet-lab experiments, initial data analysis, and manuscript drafting/editing

EDUCATION

University of Rochester

Rochester, NY

Bachelor of Science in Computer Science; Minor in Mathematics and Business

Aug 2019 – May 2023

- Relevant Coursework: Artificial Intelligence, Natural Language Processing, Deep Learning, Machine Vision

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, Rust, Java, Kotlin, SQL, C, C++

AI/ML: LLM orchestration, agent workflows, context/prompt design, LLM evals, RAG, synthetic data, vector databases, PyTorch, HuggingFace, scikit-learn, OpenCV, NumPy, Pandas

Agent / LLM Tools: Claude Code, Codex CLI, OpenCode, MCP, FastMCP, prompt/skill packages, model routing, tool use, workflow replay

Backend / Data: FastAPI, Flask, PostgreSQL, SQLite, DynamoDB, InfluxDB, MongoDB, REST APIs, WebSockets, CI/CD, Docker

Frontend / Mobile: React, TypeScript, CodeMirror 6, IndexedDB, Dexie.js, HTML/CSS, Android, Flutter

Cloud / Tools: AWS, GitLab CI/CD, GitHub Actions, Tailscale, Linux, Git, Auth0